BUSITEMA UNIVERSITY
Pursuing excellence

NABUKALU ANGEL

BU/UP/2025/2127

WAR 2

1. LOCATION

District: Kaabong District, Northern Region, Uganda - Karamoja Sub-region

Elevation: 1,508 m (4,948 ft) above sea level

County: Dodoth County, Kaabong West Municipality

Climate Zone: Aw - Tropical Savanna, semi-arid. Average temp 24°C, peak rains July.

Distance: ~400 km North of Kampala, bordering South Sudan and Kenya

2. CATCHMENT / DRAINAGE

Major System: All rivers are part of the Nile Basin - water ends in River Nile then Mediterranean Sea. Local names: Mto Kapeta, Mto Kalabi, and Loyee stream (3°44'19"N, 33°33'54"E).

Catchment Characteristics:

Soils are shallow, high runoff, many valley tanks and dams - e.g. Longoromit Dam (1.4M liters) in Lobongia Sub-County, which silts quickly due to cultivation around catchment

Main water sources: Boreholes 65-96%, dams, valley tanks, natural ponds

3. STATISTICAL SUMMARY

Year | Annual Total (mm)

2016 | 702.00

2017 | 802.72

2018 | 1007.68

2019 | 883.93

2020 | 1228.73 (Wettest)

2021 | 618.22 (Driest)

2022 | 623.35

2023 | 634.78

2024 | 974.60

2025 | 982.69

Mean Annual Rainfall: 845.87 mm

Standard Deviation: 204.93 mm* - This shows HIGH variability (24.2% coefficient of variation). Rainfall is unreliable year-to-year, typical of Karamoja.

Mean Monthly: Peak is April-August. Your data shows April 202.32mm in 2016, July 273.26mm in 2017, July 247.10mm in 2024.

4. TREND ANALYSIS

Trend (Slope) is +2.86 mm/year

Direction: Increasing - but very weak increase

R-squared is 0.0018 - This is almost 0. Meaning the trend line explains only 0.18% of the variation. Rainfall is dominated by inter-annual variability, not a clear climate change trend in this 10-year window.

Kaabong shows high inter-annual variability with no strong trend. 2020 was anomalously wet (1228mm), 2021-2023 were drought years (618-634mm). This matches the Karamoja drought bulletins.

Conclusion:

Kaabong is a semi-arid catchment in the Nile basin with highly variable rainfall (845.87 ± 204.93 mm). The catchment relies on seasonal streams and man-made dams which dry up in dry years. Over 2016-2025 there is a slight increasing trend (+2.86 mm/yr) but it is not significant (R^2=0.0018), indicating rainfall is still unpredictable and water harvesting is critical.